



MASENO UNIVERSITY
UNIVERSITY EXAMINATIONS 2024/2025

**THIRD YEAR FIRST SEMESTER EXAMINATIONS FOR THE
DEGREE OF BACHELOR OF SCIENCE IN AGRONOMY AND
BACHELOR OF SCIENCE IN ANIMAL SCIENCE WITH
INFORMATION TECHNOLOGY**

MAIN CAMPUS

AAG 307/AAB 308/ AAN 305: FUNDAMENTALS OF RESEARCH

Date: 31st January, 2025

Time: 3. 30 - 6. 30 pm

INSTRUCTIONS:

- DO NOT write anywhere on this question paper
- Answer ALL questions in SECTION A and any other TWO from SECTION B



AAG307/AAB308/AAN305: FUNDAMENTALS OF RESEARCH

Instructions: Answer **ALL** questions in section A and any **Two** selected from section B

Duration: 3 hours

Section A (Answer ALL questions)

1. What are the characteristics of a good scientist? [3 marks]
2. Explain the meaning of sampling fraction [2 marks].
3. What are the key features of the following types of research?
(a) Applied research [2 marks]
(b) Fundamental research [2 marks]
4. Using examples, explain the difference between probability and non-probability sampling methods. [4 marks]
5. Explain the difference between primary and secondary data. [2 marks].
6. Using illustrations differentiate between monotonic and non-monotonic data [4 marks]
7. Explain situations where you can use regression analysis [2 marks]
8. Explain the difference between chi square and t-tests [2 marks]
9. What are the assumptions based on linear regression analysis? [4 marks]

10. Explain the basic assumptions in scientific research [3 marks]

SECTION B: Essay Questions
(Answer Any TWO)

1. Discuss the various methods of data collection in research. [20 marks]

2. (a) Discuss the components of a research report. [12 marks]

(b) If we were to self some plants with normal wild-type phenotype of blue, large petals and observed the following number of phenotypes in the progeny:

Blue, large	182
Blue, small	60
White, large	57
White, small	21

Use Chi-square (χ^2) analysis to decide if the results fit a 9:3:3:1 ratio. (8 marks).

3. (a) Compare and contrast Simple Random sampling and Cluster sampling techniques. [10 marks]

(b) Assuming you are sampling from a population of 20,000 farmers, using Yamane's formula, calculate the representative sample size you need to get results with 95% confidence level with 5% precision. [4 marks]

(c) Cochran's formula for calculating a representative sample of a population is given as $n_o = Z^2 pq / e^2$, where n_o is the sample size, z is the selected critical value of desired confidence level, p is the estimated proportion of the attribute present in the population, $q = 1 - p$ and e is the desired precision. Assuming a large population with maximum variability, calculate the

sample sizes required that will give 95% confidence at 5% precision, and that gives 99% confidence at 1% precision. [6 marks]

4. A researcher wants to know if voting preferences (party A, party B, or party C) and gender (male, female) are related. Apply a Chi-square test to the following set of data:

- Male: Party A - 30, Party B - 20, Party C - 50
- Female: Party A - 40, Party B - 30, Party C - 30

- (a) Formulate the null and corresponding alternate hypothesis [2 marks]
- (b) Using Chi square test of independence, determine if gender does not or influences voting preferences. [8 marks]
- (c) Explain the key assumptions in chi square analysis of data. [10 marks]